2.Clustering gerarquico

November 4, 2026

[2]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import os
```

[3]:
```
data= pd.read_csv("Wholesale customers data.csv")
data.head()
```

[3]:

	Channel	Region	Fresh	Milk	Grocery	Frozen	Detergents_Paper	Delicassen
0	2	3	12669	9656	7561	214	2674	1338
1	2	3	7057	9810	9568	1762	3293	1776
2	2	3	6353	8808	7684	2405	3516	7844
3	1	3	13265	1196	4221	6404	507	1788
4	2	3	22615	5410	7198	3915	1777	5185

[4]:
```
from sklearn.preprocessing import normalize
data_scaled = normalize(data)
data_scaled
```

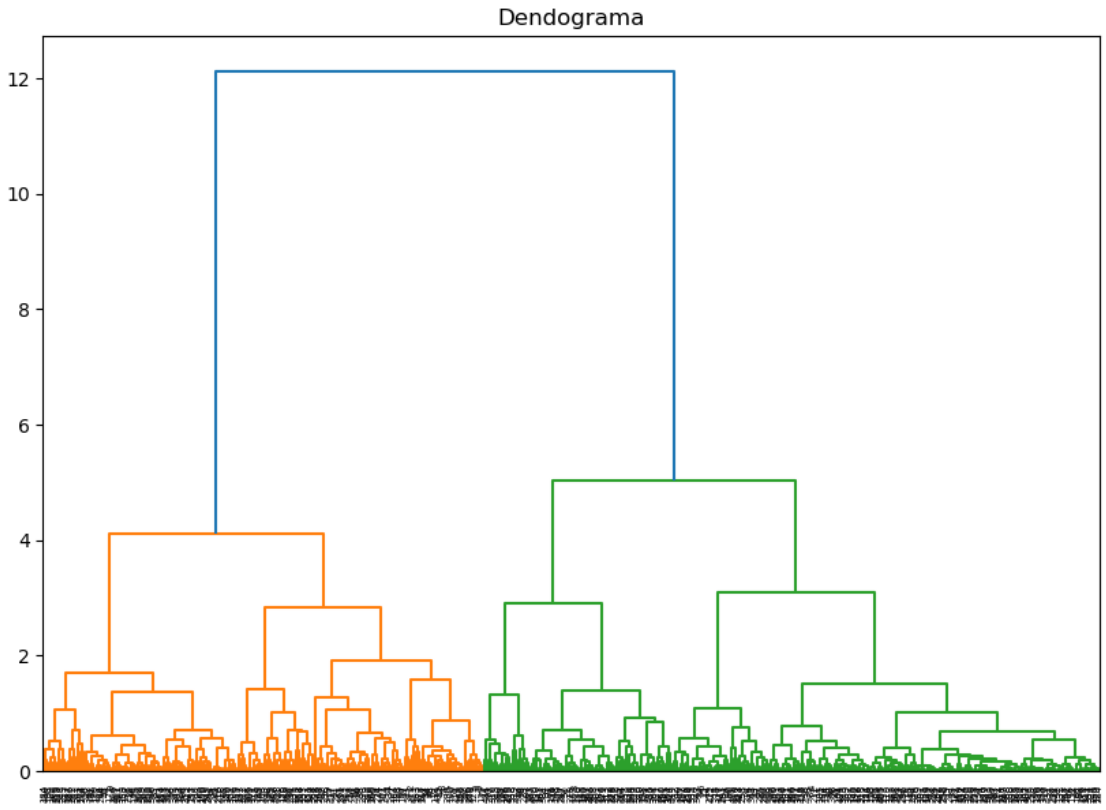
[4]:
```
array([[1.11821406e-04, 1.67732109e-04, 7.08332695e-01, …,
        1.19648904e-02, 1.49505220e-01, 7.48085205e-02],
       [1.25321880e-04, 1.87982820e-04, 4.42198253e-01, …,
        1.10408576e-01, 2.06342475e-01, 1.11285829e-01],
       [1.24839188e-04, 1.87258782e-04, 3.96551681e-01, …,
        1.50119124e-01, 2.19467293e-01, 4.89619296e-01],
       …,
       [5.01633106e-05, 7.52449659e-05, 3.64461533e-01, …,
        1.09606834e-02, 3.72236846e-01, 4.68274505e-02],
       [9.11309417e-05, 2.73392825e-04, 9.37737390e-01, …,
        9.45939175e-02, 1.53099982e-02, 1.93653251e-01],
       [2.41225630e-04, 7.23676891e-04, 6.72295832e-01, …,
        1.56796660e-02, 1.15064626e-01, 1.25437328e-02]])
```

[5]:
```
data_scaled = pd.DataFrame(data_scaled, columns = data.columns)
data_scaled.head()
```

[5]:

	Channel	Region	Fresh	Milk	Grocery	Frozen	Detergents_Paper	Delicassen
0	0.000112	0.000168	0.708333	0.539874	0.422741	0.011965	0.149505	0.074809
1	0.000125	0.000188	0.442198	0.614704	0.599540	0.110409	0.206342	0.111286
2	0.000125	0.000187	0.396552	0.549792	0.479632	0.150119	0.219467	0.489619
3	0.000065	0.000194	0.856837	0.077254	0.272650	0.413659	0.032749	0.115494
4	0.000079	0.000119	0.895416	0.214203	0.284997	0.155010	0.070358	0.205294

[6]:
```
import scipy.cluster.hierarchy as shc
plt.figure(figsize=(10,7))
plt.title("Dendograma")
dend = shc.dendrogram(shc.linkage(data_scaled, method='ward'))
```



[7]:
```
print(dend['color_list'])
```

```
['C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1',
```

```
'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1',
'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1',
'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1',
'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1',
'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1',
'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1',
'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1',
'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1',
'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1',
'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1',
'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1',
'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2',
'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2',
'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2',
'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2',
'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2',
'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2',
'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2',
'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2',
'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2',
'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2',
'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2',
'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2',
'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2',
'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2',
'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2',
'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2',
'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2',
'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2',
'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2',
'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2',
'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2',
'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2',
'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2',
'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2',
'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2',
'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2',
'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2',
'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2',
'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2',
'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2',
'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C2', 'C0']
```

[8]:
```
colores_unicos = set(dend['color_list'])
colores_unicos
```

[8]:
```
{'C0', 'C1', 'C2'}
```

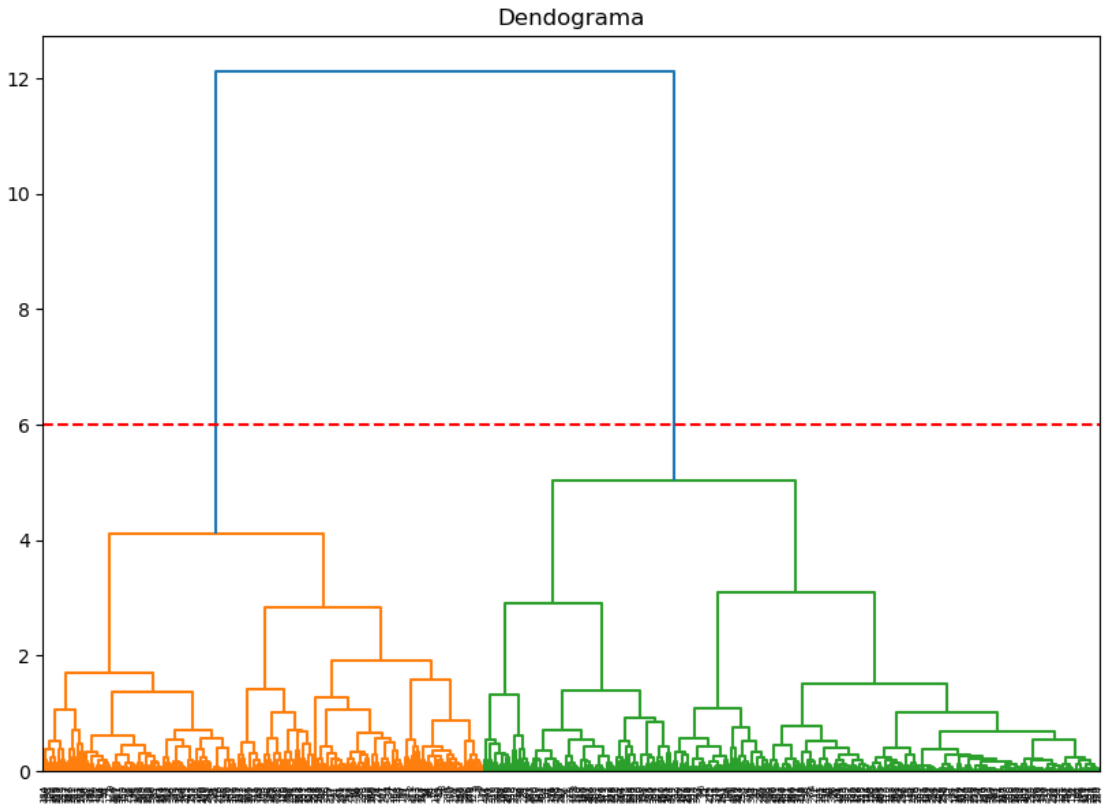
[9]:
```
num_clusters_optimo = len(colores_unicos) - 1
num_clusters_optimo
```

[9]:
```
2
```

[10]:
```
plt.figure(figsize=(10,7))
plt.title("Dendograma")
dend = shc.dendrogram(shc.linkage(data_scaled, method = 'ward'))
```

```
plt.axhline(y=6, color = 'r', linestyle = '--')
```

[10]: <matplotlib.lines.Line2D at 0x735f2bebae70>



[11]:
```
data
```

[11]:

	Channel	Region	Fresh	Milk	Grocery	Frozen	Detergents_Paper	\
0	2	3	12669	9656	7561	214	2674	
1	2	3	7057	9810	9568	1762	3293	
2	2	3	6353	8808	7684	2405	3516	
3	1	3	13265	1196	4221	6404	507	
4	2	3	22615	5410	7198	3915	1777	
..	…	…	…	…	…	…	…	
435	1	3	29703	12051	16027	13135	182	
436	1	3	39228	1431	764	4510	93	
437	2	3	14531	15488	30243	437	14841	
438	1	3	10290	1981	2232	1038	168	
439	1	3	2787	1698	2510	65	477	

	Delicassen
0	1338

```
1          1776
2          7844
3          1788
4          5185
..          …
435        2204
436        2346
437        1867
438        2125
439          52

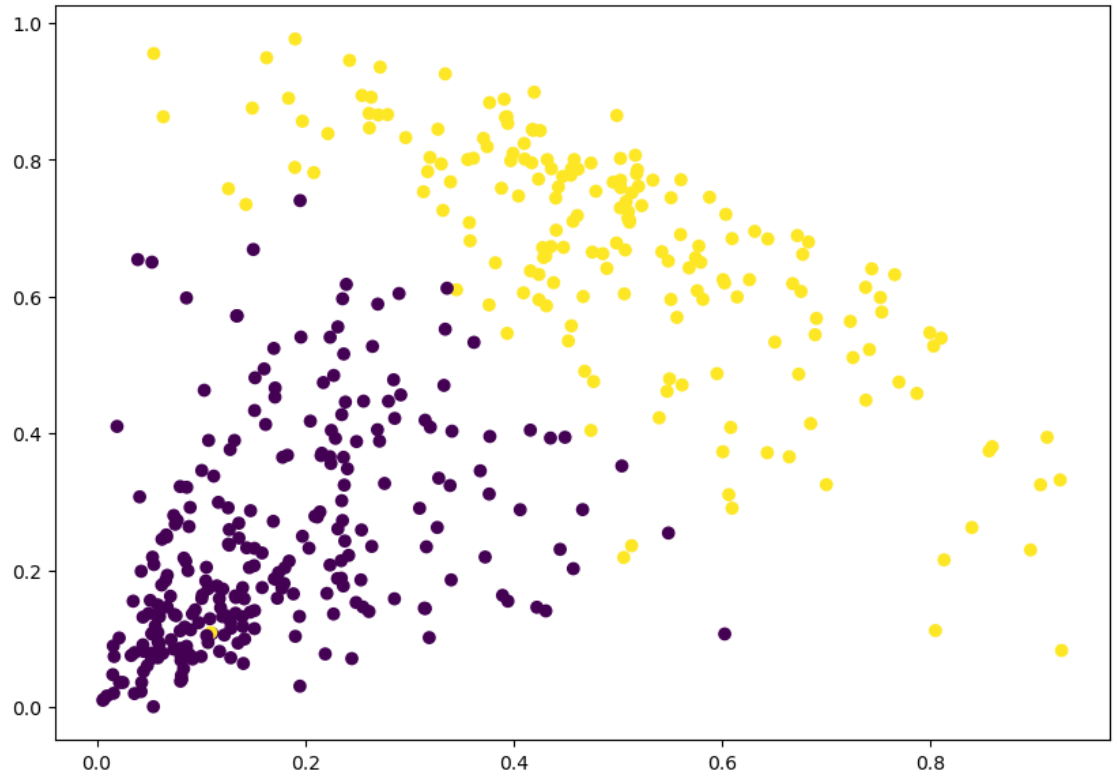
[440 rows x 8 columns]
```

[13]:
```
from sklearn.cluster import AgglomerativeClustering
cluster = AgglomerativeClustering(n_clusters=2, metric='euclidean',␣
 ↪linkage='ward')
grupos = cluster.fit_predict(data_scaled)
grupos
```

[13]:
```
array([1, 1, 1, 0, 0, 1, 0, 1, 1, 1, 1, 0, 0, 0, 0, 0, 1, 1, 0, 0, 0, 0,
       0, 1, 0, 0, 0, 0, 1, 0, 0, 1, 0, 0, 1, 1, 0, 1, 1, 0, 0, 0, 1, 1,
       1, 1, 1, 1, 1, 1, 0, 1, 0, 1, 0, 1, 1, 1, 0, 1, 0, 1, 1, 1, 0, 1,
       1, 0, 1, 0, 0, 0, 0, 0, 1, 0, 0, 1, 0, 1, 0, 1, 1, 0, 0, 1, 1, 0,
       0, 0, 0, 0, 1, 0, 1, 1, 1, 0, 0, 0, 1, 1, 1, 0, 0, 0, 1, 1, 1, 1,
       0, 1, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 1, 0, 0, 0,
       0, 0, 0, 0, 1, 1, 0, 1, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 1, 0, 1,
       0, 1, 1, 0, 1, 1, 1, 0, 0, 1, 1, 1, 1, 1, 0, 0, 1, 1, 1, 1, 1, 1,
       0, 0, 0, 1, 0, 0, 1, 1, 1, 0, 0, 1, 1, 1, 0, 0, 0, 1, 0, 0, 0, 1,
       0, 0, 1, 1, 0, 1, 1, 1, 0, 1, 1, 1, 0, 1, 0, 1, 1, 1, 1, 0, 1, 0,
       0, 1, 0, 0, 0, 0, 0, 0, 1, 0, 0, 1, 0, 1, 0, 1, 0, 0, 0, 0, 0, 0,
       0, 0, 1, 1, 0, 0, 0, 0, 0, 1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 1,
       1, 1, 1, 0, 1, 0, 0, 1, 1, 0, 1, 1, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0,
       0, 0, 0, 0, 1, 0, 1, 1, 0, 0, 0, 0, 1, 1, 0, 1, 1, 1, 1, 1, 1, 0,
       0, 1, 0, 0, 1, 0, 0, 1, 0, 1, 0, 1, 1, 0, 0, 0, 0, 0, 0, 1, 0, 0,
       0, 1, 0, 1, 0, 0, 0, 0, 0, 0, 1, 1, 1, 1, 0, 1, 1, 0, 1, 1, 0, 1,
       1, 1, 0, 1, 0, 1, 1, 1, 0, 0, 1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 1, 0,
       0, 0, 1, 0, 1, 1, 0, 0, 0, 0, 1, 0, 1, 0, 0, 0, 0, 1, 1, 0, 0, 0,
       1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 1, 1, 0, 0, 1, 1, 1,
       1, 0, 1, 0, 0, 0, 0, 0, 1, 0, 1, 0, 1, 0, 0, 1, 0, 0, 0, 1, 0, 1])
```

[14]:
```
plt.figure(figsize = (10,7))
plt.scatter(data_scaled['Milk'], data_scaled['Grocery'], c = cluster.labels_)
```

[14]:
```
<matplotlib.collections.PathCollection at 0x735f2d055c70>
```



[15]:
```
# Separación de grupos mediante PCA
campos = data_scaled.values
print(campos)
```

```
[[1.11821406e-04 1.67732109e-04 7.08332695e-01 … 1.19648904e-02
  1.49505220e-01 7.48085205e-02]
 [1.25321880e-04 1.87982820e-04 4.42198253e-01 … 1.10408576e-01
  2.06342475e-01 1.11285829e-01]
 [1.24839188e-04 1.87258782e-04 3.96551681e-01 … 1.50119124e-01
  2.19467293e-01 4.89619296e-01]
 …
 [5.01633106e-05 7.52449659e-05 3.64461533e-01 … 1.09606834e-02
  3.72236846e-01 4.68274505e-02]
 [9.11309417e-05 2.73392825e-04 9.37737390e-01 … 9.45939175e-02
  1.53099982e-02 1.93653251e-01]
 [2.41225630e-04 7.23676891e-04 6.72295832e-01 … 1.56796660e-02
  1.15064626e-01 1.25437328e-02]]
```

[16]:
```
from sklearn import decomposition
pca = decomposition.PCA(n_components = 2)
pca.fit(campos)
campos = pca.transform(campos)
```

```
campos
```

[16]:
```
array([[-3.40370496e-02, -1.80420878e-01],
       [-3.51057368e-01, -4.66441613e-02],
       [-3.05375430e-01,  5.55510662e-02],
       [ 3.83677396e-01,  1.61789121e-01],
       [ 3.18591263e-01, -7.69700156e-02],
       [-3.73478459e-02, -1.19851588e-01],
       [ 1.20176243e-01, -2.43275779e-01],
       [-2.39772413e-01, -1.04149634e-01],
       [-1.65698845e-01, -1.87946102e-01],
       [-5.68883753e-01, -1.09187885e-01],
       [-5.50490278e-01,  8.66514506e-02],
       [ 3.80404330e-01, -1.63301699e-01],
       [ 2.16997115e-01, -2.19645753e-01],
       [ 4.15793549e-02, -1.73397544e-01],
       [ 1.33933983e-01, -2.36377213e-01],
       [ 3.35298752e-01, -2.30915370e-01],
       [-7.26283916e-01, -7.33583822e-02],
       [-5.14896222e-02,  7.05034152e-03],
       [ 1.39371504e-01, -1.51057486e-01],
       [-1.64608998e-01, -2.12412480e-01],
       [ 3.33835807e-01, -1.80358027e-01],
       [ 3.23995148e-01,  2.39147842e-01],
       [ 4.98705749e-01,  4.14682390e-02],
       [-2.12232953e-01,  5.98882720e-03],
       [ 6.30009465e-02, -1.35347961e-01],
       [ 1.79665317e-01, -2.61548271e-01],
       [ 4.14823639e-01,  3.42279073e-02],
       [ 4.79249014e-01, -2.07953429e-01],
       [-6.73129427e-01, -4.79248386e-02],
       [ 5.70961913e-01, -1.91841970e-01],
       [ 1.67202588e-01, -2.10793120e-01],
       [-2.95611870e-01,  2.02389171e-01],
       [ 5.17167853e-01, -2.21203305e-01],
       [ 4.24196938e-01, -5.27010145e-02],
       [-3.54771268e-01, -3.65412400e-02],
       [-7.27927338e-01, -5.33108689e-02],
       [ 4.52523392e-01, -1.65167176e-01],
       [-1.29597534e-01, -1.53518969e-01],
       [-6.18731741e-01, -9.98567672e-02],
       [ 6.19228600e-01, -4.13006443e-02],
       [ 4.13793874e-01,  1.28551576e-01],
       [ 3.43063809e-01, -1.56418846e-01],
       [-3.07519786e-01, -2.28921480e-01],
       [-7.53223768e-01, -8.39338939e-02],
       [-2.24695460e-01, -2.01776979e-01],
```

```
[-6.22063312e-01, -1.57586831e-02],
[-6.79846268e-01, -4.73480600e-02],
[-3.27795558e-01, -9.55443701e-02],
[-1.44410163e-01, -2.16552831e-01],
[-6.74077008e-01, -6.84096223e-02],
[ 3.15602398e-01,  2.61471315e-01],
[-4.43127098e-01, -1.51451615e-01],
[ 4.90754880e-01, -2.21463763e-01],
[-7.43017325e-01, -2.17414648e-02],
[ 5.70637048e-01, -1.32305756e-01],
[-1.61897456e-01, -1.02412575e-01],
[-6.80069130e-01, -1.66615640e-02],
[-5.03231325e-01, -1.48638052e-01],
[ 5.01834682e-01, -1.78653901e-01],
[-3.05604240e-01, -1.91464732e-01],
[-1.00890901e-01, -2.67697600e-01],
[-3.99561074e-01, -1.65831759e-01],
[-1.40236863e-01,  8.28181774e-02],
[-4.08870795e-01, -7.09228640e-02],
[ 9.03889130e-02,  2.64962008e-01],
[-7.69990379e-01, -1.09298565e-01],
[-7.35890990e-01, -1.16366144e-01],
[ 4.81786081e-02, -2.21148988e-01],
[-3.25118853e-01,  4.99184692e-01],
[ 3.82260364e-01, -1.29126685e-01],
[ 4.34980462e-01,  3.00208993e-01],
[-7.45489538e-02, -4.11025454e-02],
[-4.17519668e-02,  6.42719663e-01],
[ 2.62294212e-01,  8.45335699e-02],
[-2.26029453e-01, -1.18599620e-01],
[ 6.11228251e-01, -4.80484709e-04],
[-4.29197018e-01,  6.15604204e-01],
[-4.81706738e-01, -1.79409872e-01],
[ 4.75041077e-01, -6.15378670e-02],
[-4.76281160e-01,  4.24517554e-02],
[ 2.98067142e-01, -2.11883345e-01],
[-7.67546282e-01, -7.70688157e-02],
[-2.30031687e-01, -1.54241186e-01],
[ 5.50753086e-01, -1.17615669e-01],
[ 1.94789047e-01, -1.90991741e-01],
[-6.64313844e-01, -1.34824110e-01],
[-4.54220988e-01, -2.80065613e-02],
[ 4.38157051e-01, -7.06239740e-02],
[ 2.33191454e-01,  5.23510639e-01],
[ 4.38507986e-01, -1.00163161e-01],
[ 5.27205290e-01,  3.63432139e-02],
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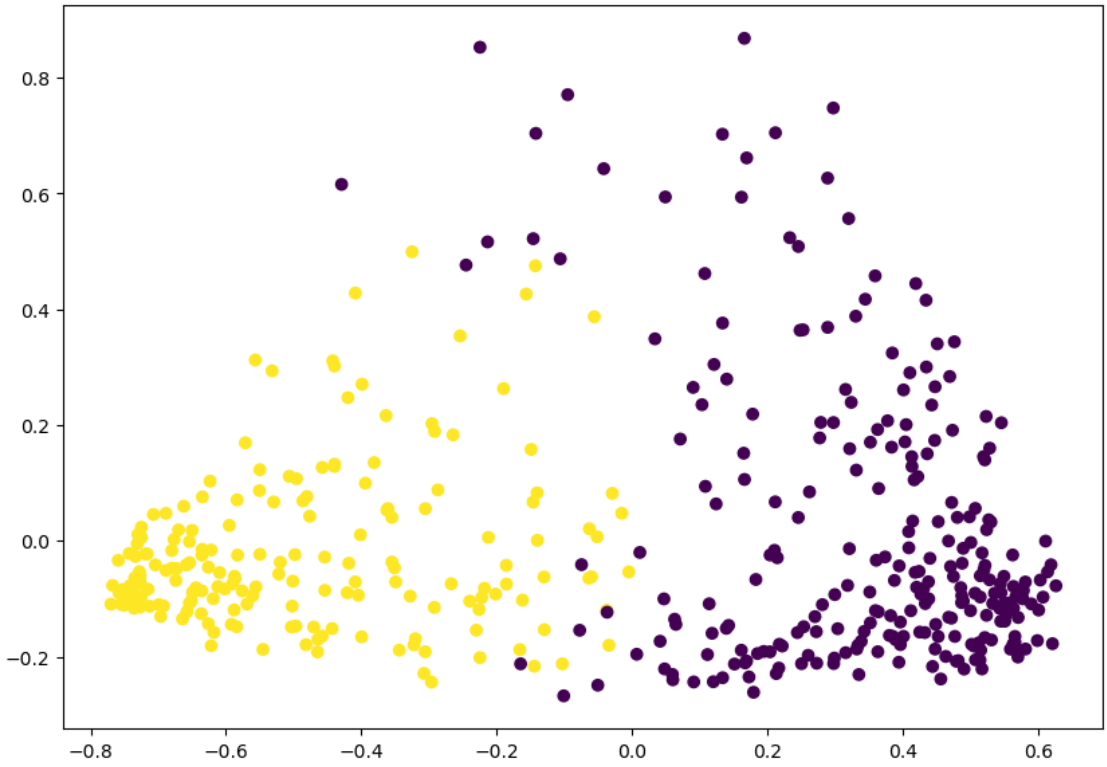
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[17]:
```
plt.figure(figsize=(10,7))
plt.scatter(campos[:,0], campos[:,1], c = cluster.labels_)
```

[17]:
```
<matplotlib.collections.PathCollection at 0x735f2bef9880>
```



[18]:
```
dataframe = pd.DataFrame(grupos, columns=['grupo'])
dataframe
```

[18]:
```
     grupo
0        1
1        1
2        1
3        0
4        0
..     …
435      0
436      0
437      1
438      0
439      1

[440 rows x 1 columns]
```

[19]:
```
dataframe2 = pd.concat([data, dataframe], axis=1, join = 'inner')
dataframe2
```

[19]:
```
     Channel  Region  Fresh   Milk  Grocery  Frozen  Detergents_Paper  \
0          2       3  12669   9656     7561     214              2674
1          2       3   7057   9810     9568    1762              3293
2          2       3   6353   8808     7684    2405              3516
3          1       3  13265   1196     4221    6404               507
4          2       3  22615   5410     7198    3915              1777
..       …     …    …    …     …       …     …                 …
435        1       3  29703  12051    16027   13135               182
436        1       3  39228   1431      764    4510                93
437        2       3  14531  15488    30243     437             14841
438        1       3  10290   1981     2232    1038               168
439        1       3   2787   1698     2510      65               477

     Delicassen  grupo
0          1338      1
1          1776      1
2          7844      1
3          1788      0
4          5185      0
..          …    …
435        2204      0
436        2346      0
437        1867      1
438        2125      0
```

```
439          52      1

[440 rows x 9 columns]
```

Analisis del caso de estudiadntes

[20]:
```
data = pd.DataFrame(np.array([[7, 6.5, 9.2, 8.6, 8],
                              [7.5, 9.4, 7.3, 7, 7],
                              [7.6, 9.2, 8, 8, 7.5],
                              [5, 6.5, 6.5, 7, 9],
                              [6, 6, 7.8, 8.9, 7.3],
                              [7.8, 9.6, 7.7, 8, 6.5],
                              [6.3, 6.4, 8.2, 9, 7.2],
                              [7.9, 9.7, 7.5, 8, 6],
                              [6, 6, 6.5, 5.5, 8.7],
                              [6.8, 7.2, 8.7, 9, 7]]),
                    columns=['Matematicas', 'Ciencias', 'Español', 'Historia',␣
 ↪'EdFisica'])
data
```

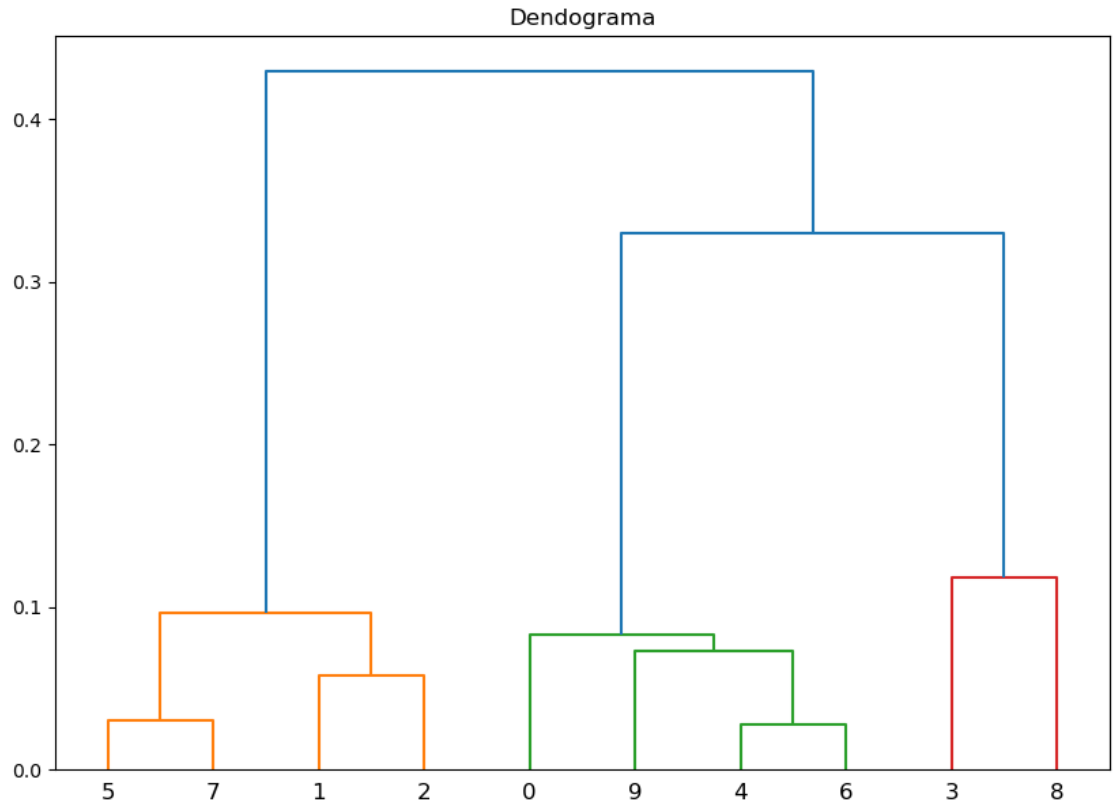
[20]:

	Matematicas	Ciencias	Español	Historia	EdFisica
0	7.0	6.5	9.2	8.6	8.0
1	7.5	9.4	7.3	7.0	7.0
2	7.6	9.2	8.0	8.0	7.5
3	5.0	6.5	6.5	7.0	9.0
4	6.0	6.0	7.8	8.9	7.3
5	7.8	9.6	7.7	8.0	6.5
6	6.3	6.4	8.2	9.0	7.2
7	7.9	9.7	7.5	8.0	6.0
8	6.0	6.0	6.5	5.5	8.7
9	6.8	7.2	8.7	9.0	7.0

[21]:
```
from sklearn.preprocessing import normalize
data_scaled = normalize(data)
data_scaled
```

[21]:
```
array([[0.39512724, 0.36690387, 0.51931009, 0.48544204, 0.45157399],
       [0.4360023 , 0.54645622, 0.42437557, 0.40693548, 0.40693548],
       [0.42050606, 0.50903365, 0.44263796, 0.44263796, 0.41497309],
       [0.32308533, 0.42001094, 0.42001094, 0.45231947, 0.5815536 ],
       [0.36834085, 0.36834085, 0.47884311, 0.54637226, 0.44814804],
       [0.43703137, 0.53788477, 0.43142841, 0.44823731, 0.36419281],
       [0.37600718, 0.38197555, 0.48940617, 0.53715311, 0.42972249],
       [0.44671284, 0.54849551, 0.42409446, 0.45236743, 0.33927557],
       [0.40434535, 0.40434535, 0.4380408 , 0.37064991, 0.58630076],
       [0.39015437, 0.41310462, 0.49916809, 0.51638078, 0.4016295 ]])
```

[22]:
```
import scipy.cluster.hierarchy as shc
plt.figure(figsize=(10,7))
plt.title("Dendograma")
dend = shc.dendrogram(shc.linkage(data_scaled, method='ward'))
```

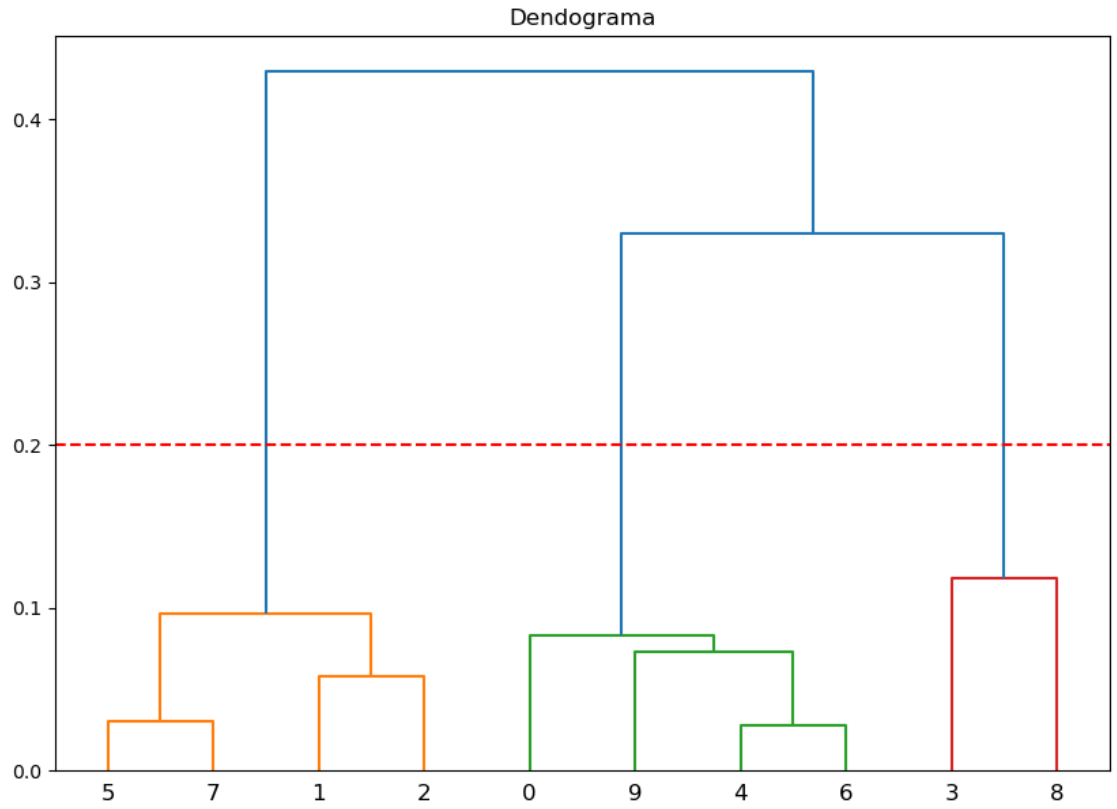


[23]:
```
colores_unicos = set(dend['color_list'])
num_clusters_optimo = len(colores_unicos)-1
num_clusters_optimo
```

[23]: 3

[24]:
```
plt.figure(figsize=(10,7))
plt.title("Dendograma")
dend = shc.dendrogram(shc.linkage(data_scaled, method = 'ward'))
plt.axhline(y=0.2, color = 'r', linestyle = '--')
```

[24]: <matplotlib.lines.Line2D at 0x735f29632c60>



[25]:
```
# Análisis gráfico con PCA
campos = data_scaled
from sklearn import decomposition
pca = decomposition.PCA(n_components = 2)
pca.fit(campos)
campos = pca.transform(campos)
campos
```

[25]:
```
array([[-0.07537384,  0.05649076],
       [ 0.11233306, -0.06156361],
       [ 0.06937082, -0.02395785],
       [-0.12493186, -0.09133353],
       [-0.08285196,  0.08431667],
       [ 0.12725973, -0.00431553],
       [-0.05987387,  0.08818057],
       [ 0.15385473,  0.00668172],
       [-0.1045932 , -0.13852098],
       [-0.01519363,  0.08402179]])
```

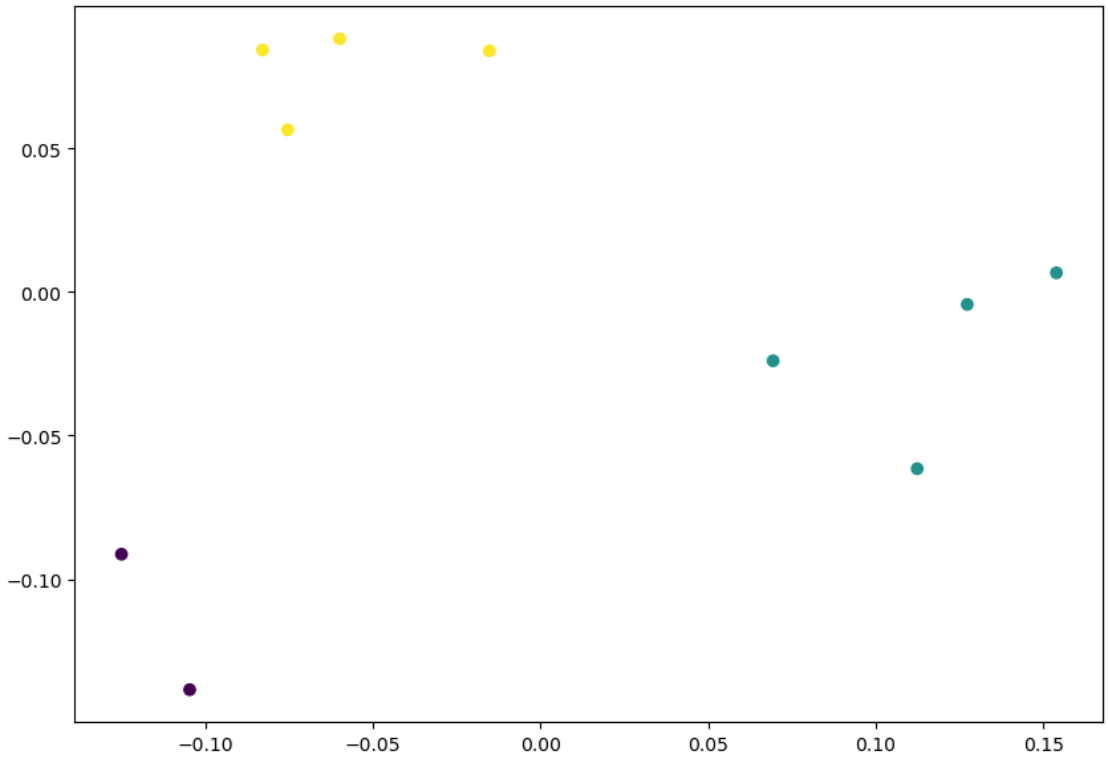
[27]:
```
from sklearn.cluster import AgglomerativeClustering
```

```
cluster = AgglomerativeClustering(n_clusters=3, metric='euclidean',␣
 ↪linkage='ward')
grupos = cluster.fit_predict(data_scaled)
grupos
```

[27]:
```
array([2, 1, 1, 0, 2, 1, 2, 1, 0, 2])
```

[28]:
```
plt.figure(figsize=(10, 7))
plt.scatter(campos[:,0], campos[:,1], c=cluster.labels_)
```

[28]:
```
<matplotlib.collections.PathCollection at 0x735f29577230>
```



[29]:
```
dataframe = pd.DataFrame(grupos, columns=['grupo'])
dataframe
```

[29]:

	grupo
0	2
1	1
2	1
3	0
4	2
5	1

```
6      2
7      1
8      0
9      2
```

[30]:
```
dataframe2 = pd.concat([data, dataframe], axis=1, join = 'inner')
dataframe2
```

[30]:

	Matematicas	Ciencias	Español	Historia	EdFisica	grupo
0	7.0	6.5	9.2	8.6	8.0	2
1	7.5	9.4	7.3	7.0	7.0	1
2	7.6	9.2	8.0	8.0	7.5	1
3	5.0	6.5	6.5	7.0	9.0	0
4	6.0	6.0	7.8	8.9	7.3	2
5	7.8	9.6	7.7	8.0	6.5	1
6	6.3	6.4	8.2	9.0	7.2	2
7	7.9	9.7	7.5	8.0	6.0	1
8	6.0	6.0	6.5	5.5	8.7	0
9	6.8	7.2	8.7	9.0	7.0	2

[31]:
```
dataframe2.index = ['Lucia', 'Pedro', 'Ines', 'Luis', 'Andres', 'Ana',␣
 ↪'Carlos', 'Jose', 'Sonia', 'Maria']
dataframe2
```

[31]:

	Matematicas	Ciencias	Español	Historia	EdFisica	grupo
Lucia	7.0	6.5	9.2	8.6	8.0	2
Pedro	7.5	9.4	7.3	7.0	7.0	1
Ines	7.6	9.2	8.0	8.0	7.5	1
Luis	5.0	6.5	6.5	7.0	9.0	0
Andres	6.0	6.0	7.8	8.9	7.3	2
Ana	7.8	9.6	7.7	8.0	6.5	1
Carlos	6.3	6.4	8.2	9.0	7.2	2
Jose	7.9	9.7	7.5	8.0	6.0	1
Sonia	6.0	6.0	6.5	5.5	8.7	0
Maria	6.8	7.2	8.7	9.0	7.0	2

[ ]:
```

```